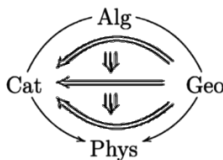


On Existence and Transport of Functorial Invariants

Math-Phys-Cat Seminars

Yuri Ximenes Martins¹

Universidade Federal de Minas Gerais
Math-Phys-Cat Group



¹webpage - nLab page

Plain

1. General Ideas
2. Formalization
3. Non-Nilpotent Invariants
4. An Obstruction Result
5. Potential Applications

This talk is based in a work in progress with Rodney Josué Biezuner.

General Ideas

Intuition

- An area of mathematics is determined by:

Intuition

- ▶ An area of mathematics is determined by:
 - ▶ the *studying objects*;

Intuition

- ▶ An area of mathematics is determined by:
 - ▶ the *studying objects*;
 - ▶ how objects *interact* with each the other.

Intuition

- ▶ An area of mathematics is determined by:
 - ▶ the *studying objects*;
 - ▶ how objects *interact* with each the other.
- ▶ When studying an area o math we think of it as something “new”. The basic problem is then the *classification problem*:

Classification Problem: Can we describe the objects of an area o math in terms of already known structures?

- This seems like a “counting” problem:

- ▶ This seems like a “counting” problem: *How many objects are there in a given math area?*
- ▶ In any counting process we have to decide if we will count or not *multiplicities*.

- ▶ This seems like a “counting” problem: *How many objects are there in a given math area?*
- ▶ In any counting process we have to decide if we will count or not *multiplicities*.
- ▶ multiplicities are components with have the same weight in the counting, meaning that they play *equivalent* role.

- ▶ This seems like a “counting” problem: *How many objects are there in a given math area?*
- ▶ In any counting process we have to decide if we will count or not *multiplicities*.
- ▶ multiplicities are components with have the same weight in the counting, meaning that they play *equivalent* role.
- ▶ the typical areas of math have such notion of “equivalence”, given by *isomorphisms*, which in turn is defined by how objects interact each other.

Classification Problem: Can we describe the objects of an area of math, *up to isomorphisms*, in terms of already known structures?

- ▶ let Ob be the “set” of objects of an area of math admitting the notion of isomorphism

Classification Problem: Are there a “set” A of “well-known” elements and a bijection as below?

$$\begin{array}{c} \text{Ob} \\ \downarrow \\ \text{Ob}/_{iso} \xrightarrow{\simeq} A \end{array}$$

- ▶ let Ob be the “set” of objects of an area of math admitting the notion of isomorphism
- ▶ let $\simeq_{iso}$ the isomorphism relation

Classification Problem: Are there a “set” A of “well-known” elements and a bijection as below?

$$\begin{array}{c} \text{Ob} \\ \downarrow \\ \text{Ob} /_{iso} \xrightarrow{\simeq} A \end{array}$$

- ▶ let Ob be the “set” of objects of an area of math admitting the notion of isomorphism
- ▶ let $\simeq_{iso}$ the isomorphism relation
- ▶ let $\text{Op} / \simeq_{iso}$ the quotient space.

Classification Problem: Are there a “set” A of “well-known” elements and a bijection as below?

$$\begin{array}{c} \text{Ob} \\ \downarrow \\ \text{Ob} /_{iso} \xrightarrow{\simeq} A \end{array}$$

Classification Problem: Are there a “set” A of “well-known” elements and a bijection as below?

$$\begin{array}{c} \text{Ob} \\ \downarrow \\ \text{Ob}/_{iso} \xrightarrow{\simeq} A \end{array}$$

- In this case, we say that the are of math is *classified* in terms of A .

Classification Problem: Are there a “set” A of “well-known” elements and a bijection as below?

$$\begin{array}{c} \text{Ob} \\ \downarrow \\ \text{Ob}/_{iso} \xrightarrow[\simeq]{} A \end{array}$$

- ▶ In this case, we say that the are of math is *classified* in terms of A .
- ▶ **Obs:** It is enough to find an injective map $F : \text{Ob}/_{iso} \rightarrow A$, since in this case F is bijective onto its image.

- ▶ The natural source of maps $F : \text{Ob} /_{iso} \rightarrow A$ are maps $f : \text{Ob} \rightarrow A$ which pass to the quotient.

- ▶ The natural source of maps $F : \text{Ob} /_{iso} \rightarrow A$ are maps $f : \text{Ob} \rightarrow A$ which pass to the quotient.

Definition: An *invariant* of a math theory is a set A and a map $f : \text{Ob} \rightarrow A$ such that there exists the dotted arrow making commutative the following diagram

$$\begin{array}{ccc} \text{Ob} & & \\ \downarrow & \searrow f & \\ \text{Ob} /_{iso} & \xrightarrow{\hat{f}} & A \end{array}$$

- ▶ Thus, an invariant assigns to each studying object $X \in \text{Ob}$ an structure/property $f(X)$ such that if $X \simeq_{iso} Y$ are isomorphic, then $f(X) = f(Y)$.

► OBS:

$$\begin{array}{ccc} \text{Ob} & \xrightarrow{s \circ F} & A \\ \downarrow & & \downarrow \uparrow s \\ \text{Ob}/_{iso} & \xrightarrow{F} & A/_{iso} \end{array}$$

► OBS:

- Typically, the well-known set A also has a notion of isomorphism.

$$\begin{array}{ccc} \text{Ob} & \xrightarrow{s \circ F} & A \\ \downarrow & & \downarrow \Big) s \\ \text{Ob}/_{iso} & \xrightarrow{F} & A/_{iso} \end{array}$$

► OBS:

- Typically, the well-known set A also has a notion of isomorphism.
- in this case, it is enough to build injective maps $F : \text{Ob} /_{iso} \rightarrow A /_{iso}$.

$$\begin{array}{ccc} \text{Ob} & \xrightarrow{s \circ F} & A \\ \downarrow & & \downarrow \uparrow s \\ \text{Ob} /_{iso} & \xrightarrow{F} & A /_{iso} \end{array}$$

► OBS:

- Typically, the well-known set A also has a notion of isomorphism.
- in this case, it is enough to build injective maps $F : \text{Ob} /_{iso} \rightarrow A /_{iso}$.
- indeed, by the Axiom of Choice the projection $A \rightarrow A /_{iso}$ has a section (which is injective too) and we can take the composition.

$$\begin{array}{ccc} \text{Ob} & \xrightarrow{s \circ F} & A \\ \downarrow & & \downarrow \uparrow s \\ \text{Ob} /_{iso} & \xrightarrow{F} & A /_{iso} \end{array}$$

► **OBS:**

► **OBS:**

- Discussion above reveals that it is enough to look at invariants $f : \text{Ob} \rightarrow A_{/iso}$

► OBS:

- Discussion above reveals that it is enough to look at invariants $f : \text{Ob} \rightarrow A/\text{iso}$
- but composing again with the section s we get a map $s \circ \hat{f} \circ \pi = s \circ f$ making commutative the diagram below.

$$\begin{array}{ccc} \text{Ob} & \xrightarrow{s \circ \hat{f}} & A \\ \pi \downarrow & \searrow f & \downarrow \uparrow s \\ \text{Ob}/\text{iso} & \xrightarrow{\hat{f}} & A/\text{iso} \end{array}$$

Conclusion: Building invariants for an area of math in terms of sets A with a notion of isomorphism is about building maps $f : \text{Ob} \rightarrow A$ which pass to the quotient, as below.

$$\begin{array}{ccc} \text{Ob} & \xrightarrow{f} & A \\ \downarrow & & \downarrow \\ \text{Ob}/_{iso} & \xrightarrow{\hat{f}} & A/_{iso} \end{array}$$

- In other words, it is about finding maps $f : \text{Ob} \rightarrow A$ such that if $X \simeq_{iso} Y$, then $f(X) \simeq_{iso} f(Y)$.

- For a fixed math area, let Inv_A denote the “set” of all invariants taking values in a set A , i.e., the collections of all $f : \text{Ob} \rightarrow A$ for which there exists $\hat{f}$ as above.

- ▶ For a fixed math area, let Inv_A denote the “set” of all invariants taking values in a set A , i.e., the collections of all $f : \text{Ob} \rightarrow A$ for which there exists $\hat{f}$ as above.
- ▶ given two classes $\mathcal{C}_A, \mathcal{D}_A \subset \text{Inv}_A$ of invariants one can ask whether they are “independent”.

- ▶ For a fixed math area, let Inv_A denote the “set” of all invariants taking values in a set A , i.e., the collections of all $f : \text{Ob} \rightarrow A$ for which there exists $\hat{f}$ as above.
- ▶ given two classes $\mathcal{C}_A, \mathcal{D}_A \subset \text{Inv}_A$ of invariants one can ask whether they are “independent”.

Definition. We say that $\mathcal{C}_A$ and $\mathcal{D}_A$ *does not exist mutually* if $\mathcal{C}_A \neq \emptyset$ implies $\mathcal{D}_A = \emptyset$ or if $\mathcal{D}_A \neq \emptyset$ implies $\mathcal{C}_A = \emptyset$. Write $\mathcal{C}_A \# \mathcal{D}_A$.

- ▶ Given $\mathcal{C}_A$, let $\#(\mathcal{C}_A)$ be the collection of every $\mathcal{D}_A$ such that $\mathcal{C}_A \# \mathcal{D}_A$.

- Now, suppose you are trying to find some invariant satisfying specific properties. Let $\mathcal{C}_A$ be the class of all those invariants.

- ▶ Now, suppose you are trying to find some invariant satisfying specific properties. Let $\mathcal{C}_A$ be the class of all those invariants.
- ▶ If one proves that another class $\mathcal{D}_A \in \#(\mathcal{C}_A)$ is nonempty, then such a desired invariant does not exist.

- ▶ Now, suppose you are trying to find some invariant satisfying specific properties. Let $\mathcal{C}_A$ be the class of all those invariants.
- ▶ If one proves that another class $\mathcal{D}_A \in \sharp(\mathcal{C}_A)$ is nonempty, then such a desired invariant does not exist.
- ▶ Thus, *classes in $\sharp(\mathcal{C}_A)$ are obstructions to the existence of invariants belonging to $\mathcal{C}_A$.*

- ▶ Now, suppose you are trying to find some invariant satisfying specific properties. Let $\mathcal{C}_A$ be the class of all those invariants.
- ▶ If one proves that another class $\mathcal{D}_A \in \sharp(\mathcal{C}_A)$ is nonempty, then such a desired invariant does not exist.
- ▶ Thus, *classes in $\sharp(\mathcal{C}_A)$ are obstructions to the existence of invariants belonging to $\mathcal{C}_A$.*
- ▶ We proved that, if A has certain nature, then for every $\mathcal{C}_A$ there exists a class $\text{NNilp}(\mathcal{C}_A)$ of “non-nilpotent” invariants such that $\text{NNilp}(\mathcal{C}_A) \in \sharp(\mathcal{C}_A)$.

- How could this be useful?

- ▶ How could this be useful?

We say that an invariant $f : \text{Ob} \rightarrow A$ is *complete* if $\hat{f} : \text{Ob} /_{iso} \rightarrow A_{iso}$ is injective.

- ▶ Thus, an invariant is complete if it strong enough to solve the classification problem.

- ▶ How could this be useful?

We say that an invariant $f : \text{Ob} \rightarrow A$ is *complete* if $\hat{f} : \text{Ob} /_{iso} \rightarrow A_{iso}$ is injective.

- ▶ Thus, an invariant is complete if it strong enough to solve the classification problem.
- ▶ Let $\text{Comp}_A \subset \text{Inv}_A$ be the collection of all A -valued complete invariants. If A has certain nature, then Comp_A is nonempty only if there are A -valued “non-nilpotent” invariants....

- ▶ In every math area with a notion of isomorphisms we have a canonical invariant $\text{Aut} : \text{Ob} \rightarrow \text{Grp}$ assigning to each object X its automorphism group $\text{Aut}(X)$.

- ▶ In every math area with a notion of isomorphisms we have a canonical invariant $\text{Aut} : \text{Ob} \rightarrow \text{Grp}$ assigning to each object X its automorphism group $\text{Aut}(X)$.

Isomorphism Problem: Is Aut a complete invariant?

- ▶ Can a set X be reconstructed from its permutations?

- ▶ In every math area with a notion of isomorphisms we have a canonical invariant $\text{Aut} : \text{Ob} \rightarrow \text{Grp}$ assigning to each object X its automorphism group $\text{Aut}(X)$.

Isomorphism Problem: Is Aut a complete invariant?

- ▶ Can a set X be reconstructed from its permutations?
- ▶ Can a ring R be reconstructed from isomorphism group $\text{Iso}(R)$?

- ▶ In every math area with a notion of isomorphisms we have a canonical invariant $\text{Aut} : \text{Ob} \rightarrow \text{Grp}$ assigning to each object X its automorphism group $\text{Aut}(X)$.

Isomorphism Problem: Is Aut a complete invariant?

- ▶ Can a set X be reconstructed from its permutations?
- ▶ Can a ring R be reconstructed from isomorphism group $\text{Iso}(R)$?
- ▶ Can a topological space X be reconstructed from its homeomorphism group $\text{Homeo}(X)$;

- ▶ In every math area with a notion of isomorphisms we have a canonical invariant $\text{Aut} : \text{Ob} \rightarrow \text{Grp}$ assigning to each object X its automorphism group $\text{Aut}(X)$.

Isomorphism Problem: Is Aut a complete invariant?

- ▶ Can a set X be reconstructed from its permutations?
- ▶ Can a ring R be reconstructed from isomorphism group $\text{Iso}(R)$?
- ▶ Can a topological space X be reconstructed from its homeomorphism group $\text{Homeo}(X)$;
- ▶ Can a smooth manifold M be reconstructed from its diffeomorphism group $\text{Diff}(M)$ (...)

Formalization

- ▶ the typical context in which we have objects, interactions between objects and isomorphisms is a category.

- ▶ the typical context in which we have objects, interactions between objects and isomorphisms is a category.

An area of math *is* a category $\mathbf{C}$.

- ▶ $\text{Ob}(\mathbf{C})$ is the collection of studying objects

- ▶ the typical context in which we have objects, interactions between objects and isomorphisms is a category.

An area of math *is* a category $\mathbf{C}$.

- ▶ $\text{Ob}(\mathbf{C})$ is the collection of studying objects
- ▶ a relation between X and Y is a morphism $f : X \rightarrow Y$

- ▶ the typical context in which we have objects, interactions between objects and isomorphisms is a category.

An area of math *is* a category $\mathbf{C}$.

- ▶ $\text{Ob}(\mathbf{C})$ is the collection of studying objects
- ▶ a relation between X and Y is a morphism $f : X \rightarrow Y$
- ▶ $X \simeq_{iso} Y$ if there is an isomorphism $f : X \rightarrow Y$.

- Functors $F : \mathbf{C} \rightarrow \mathbf{A}$ preserve isomorphisms, so that its function on objects pass to the quotient, as below.

- Functors $F : \mathbf{C} \rightarrow \mathbf{A}$ preserve isomorphisms, so that its function on objects pass to the quotient, as below.

$$\begin{array}{ccc}
 \mathrm{Ob}(\mathbf{C}) & \xrightarrow{f} & \mathrm{Ob}(\mathbf{D}) \\
 \downarrow & & \downarrow \\
 \mathrm{Ob}(\mathbf{C})/_{iso} & \xrightarrow[\hat{f}]{} & \mathrm{Ob}(\mathbf{D})/_{iso}
 \end{array}$$

- ▶ Functors $F : \mathbf{C} \rightarrow \mathbf{A}$ preserve isomorphisms, so that its function on objects pass to the quotient, as below.

$$\begin{array}{ccc}
 \mathrm{Ob}(\mathbf{C}) & \xrightarrow{f} & \mathrm{Ob}(\mathbf{D}) \\
 \downarrow & & \downarrow \\
 \mathrm{Ob}(\mathbf{C})/_{iso} & \xrightarrow[\hat{f}]{} & \mathrm{Ob}(\mathbf{D})/_{iso}
 \end{array}$$

Every functor $F : \mathbf{C} \rightarrow \mathbf{A}$ defines an invariant in the math area defined by $\mathbf{C}$.

- ▶ if $\mathbf{C}$ is not a groupoid, reciprocal is generally false.

Definition. An invariant of $\mathbf{C}$ which comes from a functor $F : \mathbf{C} \rightarrow \mathbf{D}$ is a *functorial $\mathbf{A}$ -invariant*.

- Let $\text{Inv}_{\mathbf{D}}(\mathbf{C})$ denote the collection of all of them.

Definition. An invariant of $\mathbf{C}$ which comes from a functor $F : \mathbf{C} \rightarrow \mathbf{D}$ is a *functorial $\mathbf{A}$ -invariant*.

- ▶ Let $\text{Inv}_{\mathbf{D}}(\mathbf{C})$ denote the collection of all of them.
- ▶ We also say that $\mathbf{D}$ is the *coefficients category*.

Definition. An invariant of $\mathbf{C}$ which comes from a functor $F : \mathbf{C} \rightarrow \mathbf{D}$ is a *functorial $\mathbf{A}$ -invariant*.

- ▶ Let $\text{Inv}_{\mathbf{D}}(\mathbf{C})$ denote the collection of all of them.
- ▶ We also say that $\mathbf{D}$ is the *coefficients category*.
- ▶ In $\text{Inv}_{\mathbf{D}}(\mathbf{C})$ we have another relation.

Definition. An invariant of $\mathbf{C}$ which comes from a functor $F : \mathbf{C} \rightarrow \mathbf{D}$ is a *functorial $\mathbf{A}$ -invariant*.

- ▶ Let $\text{Inv}_{\mathbf{D}}(\mathbf{C})$ denote the collection of all of them.
- ▶ We also say that $\mathbf{D}$ is the *coefficients category*.
- ▶ In $\text{Inv}_{\mathbf{D}}(\mathbf{C})$ we have another relation.

Definition. A *constrain* between functorial $\mathbf{D}$ -invariants $F, F' : \mathbf{C} \rightarrow \mathbf{D}$ is a non-natural transformation $\xi : F \Rightarrow F'$, i.e., a family of morphisms $\xi_X : F(X) \rightarrow F'(X)$, with $X \in \text{Ob}(\mathbf{C})$.

Definition. A *strong constrain* (or *non-natural isomorphism*) between $F, F' : \mathbf{C} \rightarrow \mathbf{D}$ is a constrain $\xi : F \Rightarrow F'$ whose morphisms $\xi_X : F(X) \rightarrow F'(X)$ are isomorphisms for every X . We write $F \approx F'$.

- Sometimes we build functorial $\mathbf{D}$ -invariants with $\mathbf{D}$ poor of properties. In this case, we would like to “transport” our invariant to other more well-behaved coefficient category.

Definition. A *strong constrain* (or *non-natural isomorphism*) between $F, F' : \mathbf{C} \rightarrow \mathbf{D}$ is a constrain $\xi : F \Rightarrow F'$ whose morphisms $\xi_X : F(X) \rightarrow F'(X)$ are isomorphisms for every X . We write $F \approx F'$.

- ▶ Sometimes we build functorial $\mathbf{D}$ -invariants with $\mathbf{D}$ poor of properties. In this case, we would like to “transport” our invariant to other more well-behaved coefficient category.
- ▶ Composition of functors give us a map

$$\tau : \text{Inv}_{\mathbf{D}}(\mathbf{C}) \times \text{Inv}_{\mathbf{D}'}(\mathbf{D}) \rightarrow \text{Inv}_{\mathbf{D}'}(\mathbf{C}).$$

Definition. A *strong constrain* (or *non-natural isomorphism*) between $F, F' : \mathbf{C} \rightarrow \mathbf{D}$ is a constrain $\xi : F \Rightarrow F'$ whose morphisms $\xi_X : F(X) \rightarrow F'(X)$ are isomorphisms for every X . We write $F \approx F'$.

- ▶ Sometimes we build functorial $\mathbf{D}$ -invariants with $\mathbf{D}$ poor of properties. In this case, we would like to “transport” our invariant to other more well-behaved coefficient category.
- ▶ Composition of functors give us a map

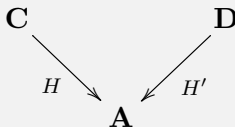
$$\tau : \text{Inv}_{\mathbf{D}}(\mathbf{C}) \times \text{Inv}_{\mathbf{D}'}(\mathbf{D}) \rightarrow \text{Inv}_{\mathbf{D}'}(\mathbf{C}).$$

Definition. We say that $\tau_T(F) \equiv \tau(F, T)$ is the *transport of F along T* .

- ▶ sometimes we consider spans of $\mathbf{A}$ -invariants, which model the idea of two analogous invariants defined in different areas of math.

- ▶ sometimes we consider spans of **A**-invariants, which model the idea of two analogous invariants defined in different areas of math.

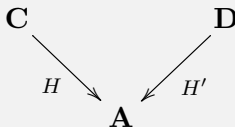
Definition. A *functorial **A**-invariant* in $(\mathbf{C}, \mathbf{D})$ is a pair (H, H') of **A**-invariants, as below



- ▶ Two pairs (H, H') and (K, K') are *non-naturally isomorphic* (or *strongly constrained*) if $H \approx K$ and $H' \approx K'$.

- ▶ sometimes we consider spans of $\mathbf{A}$ -invariants, which model the idea of two analogous invariants defined in different areas of math.

Definition. A *functorial $\mathbf{A}$ -invariant in $(\mathbf{C}, \mathbf{D})$* is a pair (H, H') of $\mathbf{A}$ -invariants, as below



- ▶ Two pairs (H, H') and (K, K') are *non-naturally isomorphic* (or *strongly constrained*) if $H \approx K$ and $H' \approx K'$.
- ▶ Let $\text{Inv}_{\mathbf{A}}(\mathbf{C}, \mathbf{D})$ be the collection of all such spans.

- We can transport invariants along invariants spans

$$\tau : \text{Inv}_{\mathbf{D}}(\mathbf{C}) \times \text{Inv}_{\mathbf{A}}(\mathbf{C}, \mathbf{D}) \rightarrow \text{Inv}_{\mathbf{A}}(\mathbf{C}) \times \text{Inv}_{\mathbf{A}}(\mathbf{C}),$$

- We can transport invariants along invariants spans

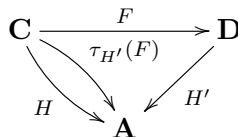
$$\tau : \text{Inv}_{\mathbf{D}}(\mathbf{C}) \times \text{Inv}_{\mathbf{A}}(\mathbf{C}, \mathbf{D}) \rightarrow \text{Inv}_{\mathbf{A}}(\mathbf{C}) \times \text{Inv}_{\mathbf{A}}(\mathbf{C}),$$

such that $\tau(F, (H, H')) = (H, \tau_{H'}(F)) = (H, H' \circ F)$.

- We can transport invariants along invariants spans

$$\tau : \text{Inv}_{\mathbf{D}}(\mathbf{C}) \times \text{Inv}_{\mathbf{A}}(\mathbf{C}, \mathbf{D}) \rightarrow \text{Inv}_{\mathbf{A}}(\mathbf{C}) \times \text{Inv}_{\mathbf{A}}(\mathbf{C}),$$

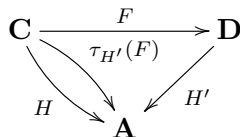
such that $\tau(F, (H, H')) = (H, \tau_{H'}(F)) = (H, H' \circ F)$.



- We can transport invariants along invariants spans

$$\tau : \text{Inv}_{\mathbf{D}}(\mathbf{C}) \times \text{Inv}_{\mathbf{A}}(\mathbf{C}, \mathbf{D}) \rightarrow \text{Inv}_{\mathbf{A}}(\mathbf{C}) \times \text{Inv}_{\mathbf{A}}(\mathbf{C}),$$

such that $\tau(F, (H, H')) = (H, \tau_{H'}(F)) = (H, H' \circ F)$.



Definition. We say that the transport $\tau_{H,H'}(F)$ of F along (H, H') is *effective* if $\tau_{H'}(F) \approx H$.

- But we also know how to transport spans of invariants along invariants:

$$\tau : \text{Inv}_{\mathbf{V}}(\mathbf{C}, \mathbf{D}) \times \text{Inv}_{\mathbf{A}}(\mathbf{V}) \rightarrow \text{Inv}_{\mathbf{V}}(\mathbf{C}, \mathbf{D}),$$

- But we also know how to transport spans of invariants along invariants:

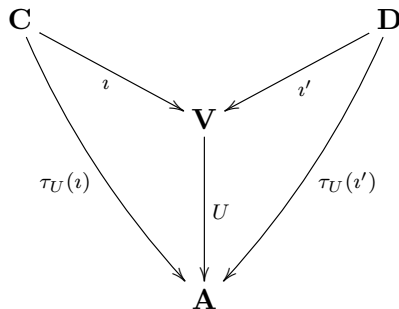
$$\tau : \text{Inv}_{\mathbf{V}}(\mathbf{C}, \mathbf{D}) \times \text{Inv}_{\mathbf{A}}(\mathbf{V}) \rightarrow \text{Inv}_{\mathbf{V}}(\mathbf{C}, \mathbf{D}),$$

such that $\tau((\iota, \iota'), U) = (\tau_U(\iota), \tau_U(\iota'))$. We write $\tau^U(\iota, \iota')$.

- But we also know how to transport spans of invariants along invariants:

$$\tau : \text{Inv}_{\mathbf{V}}(\mathbf{C}, \mathbf{D}) \times \text{Inv}_{\mathbf{A}}(\mathbf{V}) \rightarrow \text{Inv}_{\mathbf{V}}(\mathbf{C}, \mathbf{D}),$$

such that $\tau((\iota, \iota'), U) = (\tau_U(\iota), \tau_U(\iota'))$. We write $\tau^U(\iota, \iota')$.



- Thus, given F and (ι, ι') , we can take the iterated transport $\tau^U(\tau_{\iota, \iota'}(F))$, called a *double transport of F* .

- Thus, given F and $(\imath, \imath')$, we can take the iterated transport $\tau^U(\tau_{\imath, \imath'}(F))$, called a *double transport of F* .

Definition. A double transport $\tau^U(\tau_{\imath, \imath'}(F))$ of F is *effective* if it is effective as a transport of F along $(\tau_U(\imath), \tau_U(\imath'))$.

- ▶ Thus, given F and (ι, ι') , we can take the iterated transport $\tau^U(\tau_{\iota, \iota'}(F))$, called a *double transport of F* .

Definition. A double transport $\tau^U(\tau_{\iota, \iota'}(F))$ of F is *effective* if it is effective as a transport of F along $(\tau_U(\iota), \tau_U(\iota'))$.

- ▶ $\tau^U(\tau_{\iota, \iota'}(F))$ effective does not necessarily means that $\tau_{\iota, \iota'}(F)$ is effective (but holds if U is essentially injective).

Non-Nilpotent Invariants

- if $(X, *)$ is some space with a multiplication and with a zero, we say that $x \in X$ is *nilpotent* if there exists r such that $x^k = x * x * x \dots * x = 0$ for every $k > r$

- ▶ if $(X, *)$ is some space with a multiplication and with a zero, we say that $x \in X$ is *nilpotent* if there exists r such that $x^k = x * x * x \dots * x = 0$ for every $k > r$
- ▶ more generally, we say that a set $S \subset X$ is *nilpotent* if there is r such that for every $x_1, \dots, x_k \in S$, with $k > r$, we have $((x_1 * x_2) * x_3) * \dots * x_k = 0$.

- ▶ if $(X, *)$ is some space with a multiplication and with a zero, we say that $x \in X$ is *nilpotent* if there exists r such that $x^k = x * x * x \dots * x = 0$ for every $k > r$
- ▶ more generally, we say that a set $S \subset X$ is *nilpotent* if there is r such that for every $x_1, \dots, x_k \in S$, with $k > r$, we have $((x_1 * x_2) * x_3) * \dots * x_k = 0$.
- ▶ we could also try this notion for vector spaces. A collection S of vector spaces is *nilpotent* if there is k such that for every $X_1, \dots, X_k$, with $k > r$ we have $(X_1 \otimes X_2) \otimes \dots X_k \simeq \mathbb{R}$.

- more generally, given a monoidal category $(\mathbf{V}, \otimes, 1)$, we say that a collection of objects $S \subset \text{Ob}(\mathbf{V})$ is *nilpotent* if there exists r such that for every $X_1, \dots, X_k \in S$, with $k > r$, we have $(X_1 \otimes X_2) \otimes \dots X_k \simeq 1$.

- ▶ more generally, given a monoidal category $(\mathbf{V}, \otimes, 1)$, we say that a collection of objects $S \subset \text{Ob}(\mathbf{V})$ is *nilpotent* if there exists r such that for every $X_1, \dots, X_k \in S$, with $k > r$, we have $(X_1 \otimes X_2) \otimes \dots X_k \simeq 1$.
- ▶ In all these examples we say that the smaller k satisfying those conditions is the *nilpotency degree of S* .

- ▶ more generally, given a monoidal category $(\mathbf{V}, \otimes, 1)$, we say that a collection of objects $S \subset \text{Ob}(\mathbf{V})$ is *nilpotent* if there exists r such that for every $X_1, \dots, X_k \in S$, with $k > r$, we have $(X_1 \otimes X_2) \otimes \dots X_k \simeq 1$.
- ▶ In all these examples we say that the smaller k satisfying those conditions is the *nilpotency degree of S* .
- ▶ Dually, we have the notion of non-nilpotent set.

- ▶ more generally, given a monoidal category $(\mathbf{V}, \otimes, 1)$, we say that a collection of objects $S \subset \text{Ob}(\mathbf{V})$ is *nilpotent* if there exists r such that for every $X_1, \dots, X_k \in S$, with $k > r$, we have $(X_1 \otimes X_2) \otimes \dots X_k \simeq 1$.
- ▶ In all these examples we say that the smaller k satisfying those conditions is the *nilpotency degree of S* .
- ▶ Dually, we have the notion of non-nilpotent set.

Definition. We say that a collection $S \subset \text{Ob}(\mathbf{V})$ is *non-nilpotent of degree r* if there exists $X \in S$ with the following property: for every list $X_1, \dots, X_k$ of objects in $\mathbf{V}$, with $k > r$, if $X_i \simeq X$ for some i , then $(X_1 \otimes X_2) \otimes \dots \otimes X_k \neq 1$.

- ▶ When a structure is generated by a set, properties valid in the generating set typically extends to the entire structure.

- ▶ When a structure is generated by a set, properties valid in the generating set typically extends to the entire structure.

Definition. A *generating structure* $\mathcal{S}$ for a monoidal category $(\mathbf{V}, \otimes, 1)$ is given by

1. a collection S of objects of $\mathbf{V}$, called *generating set*;

- When a structure is generated by a set, properties valid in the generating set typically extends to the entire structure.

Definition. A *generating structure* $\mathcal{S}$ for a monoidal category $(\mathbf{V}, \otimes, 1)$ is given by

1. a collection S of objects of $\mathbf{V}$, called *generating set*;
2. for every object $X \in \mathbf{V}$:

- When a structure is generated by a set, properties valid in the generating set typically extends to the entire structure.

Definition. A *generating structure* $\mathcal{S}$ for a monoidal category $(\mathbf{V}, \otimes, 1)$ is given by

1. a collection S of objects of $\mathbf{V}$, called *generating set*;
2. for every object $X \in \mathbf{V}$:
 - 2.1 a finite interval $I_X \subset P$;

- When a structure is generated by a set, properties valid in the generating set typically extends to the entire structure.

Definition. A *generating structure* $\mathcal{S}$ for a monoidal category $(\mathbf{V}, \otimes, 1)$ is given by

1. a collection S of objects of $\mathbf{V}$, called *generating set*;
2. for every object $X \in \mathbf{V}$:
 - 2.1 a finite interval $I_X \subset P$;
 - 2.2 a family $S_X = (X_i)$, with $i \in I_X$ and $X_i \in S$;

- When a structure is generated by a set, properties valid in the generating set typically extends to the entire structure.

Definition. A *generating structure* $\mathcal{S}$ for a monoidal category $(\mathbf{V}, \otimes, 1)$ is given by

1. a collection S of objects of $\mathbf{V}$, called *generating set*;
2. for every object $X \in \mathbf{V}$:
 - 2.1 a finite interval $I_X \subset P$;
 - 2.2 a family $S_X = (X_i)$, with $i \in I_X$ and $X_i \in S$;
 - 2.3 an isomorphism $u_X : X \simeq \otimes_{I_X} X_i$.

- ▶ When a structure is generated by a set, properties valid in the generating set typically extends to the entire structure.

Definition. A *generating structure* $\mathcal{S}$ for a monoidal category $(\mathbf{V}, \otimes, 1)$ is given by

1. a collection S of objects of $\mathbf{V}$, called *generating set*;
 2. for every object $X \in \mathbf{V}$:
 - 2.1 a finite interval $I_X \subset P$;
 - 2.2 a family $S_X = (X_i)$, with $i \in I_X$ and $X_i \in S$;
 - 2.3 an isomorphism $u_X : X \simeq \otimes_{I_X} X_i$.
- ▶ A monoidal category is *non-nilpotent of degree r* if it admits a generating structure $\mathcal{S}$ whose generating set S is non-nilpotent of degree r .

Definition. A functor $U : (\mathbf{V}, \otimes, 1) \rightarrow (\mathbf{A}, \odot, *)$ between monoidal categories is:

1. *multiplicative* if it is a strong monoidal functor;

Definition. A functor $U : (\mathbf{V}, \otimes, 1) \rightarrow (\mathbf{A}, \odot, *)$ between monoidal categories is:

1. *multiplicative* if it is a strong monoidal functor;
2. *left cancellative* if it satisfies the following left-cancellation law: for every $A, B, C \in U(\mathbf{V})$, if $B \odot A \simeq C \odot A$, then $B \simeq C$;

Definition. A functor $U : (\mathbf{V}, \otimes, 1) \rightarrow (\mathbf{A}, \odot, *)$ between monoidal categories is:

1. *multiplicative* if it is a strong monoidal functor;
2. *left cancellative* if it satisfies the following left-cancellation law: for every $A, B, C \in U(\mathbf{V})$, if $B \odot A \simeq C \odot A$, then $B \simeq C$;
3. *coherent* if it is both multiplicative and left cancellative;

Definition. A functor $U : (\mathbf{V}, \otimes, 1) \rightarrow (\mathbf{A}, \odot, *)$ between monoidal categories is:

1. *multiplicative* if it is a strong monoidal functor;
2. *left cancellative* if it satisfies the following left-cancellation law: for every $A, B, C \in U(\mathbf{V})$, if $B \odot A \simeq C \odot A$, then $B \simeq C$;
3. *coherent* if it is both multiplicative and left cancellative;
4. *defines non-nilpotent invariants* if it is multiplicative and the monoidal subcategory $U(\mathbf{V})$ is non-nilpotent of degree $r > 1$.

An Obstruction Result

Theorem. Let $\mathbf{C}, \mathbf{D}$ categories, $(\mathbf{V}, \otimes, 1)$ and $(\mathbf{A}, \odot, *)$ monoidal categories and $U : \mathbf{V} \rightarrow \mathbf{A}$ a functor.

Theorem. Let $\mathbf{C}, \mathbf{D}$ categories, $(\mathbf{V}, \otimes, 1)$ and $(\mathbf{A}, \odot, *)$ monoidal categories and $U : \mathbf{V} \rightarrow \mathbf{A}$ a functor. Given a class $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$ of functorial $\mathbf{D}$ -invariants, the following cannot be satisfied simultaneously:

Theorem. Let $\mathbf{C}, \mathbf{D}$ categories, $(\mathbf{V}, \otimes, 1)$ and $(\mathbf{A}, \odot, *)$ monoidal categories and $U : \mathbf{V} \rightarrow \mathbf{A}$ a functor.

Given a class $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$ of functorial $\mathbf{D}$ -invariants, the following cannot be satisfied simultaneously:

- ▶ $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$ is non-empty;

Theorem. Let $\mathbf{C}, \mathbf{D}$ categories, $(\mathbf{V}, \otimes, 1)$ and $(\mathbf{A}, \odot, *)$ monoidal categories and $U : \mathbf{V} \rightarrow \mathbf{A}$ a functor.

Given a class $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$ of functorial $\mathbf{D}$ -invariants, the following cannot be satisfied simultaneously:

- ▶ $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$ is non-empty;
- ▶ U is coherent and admits a complement structure $(-)^c$ relatively to $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$

Theorem. Let $\mathbf{C}, \mathbf{D}$ categories, $(\mathbf{V}, \otimes, 1)$ and $(\mathbf{A}, \odot, *)$ monoidal categories and $U : \mathbf{V} \rightarrow \mathbf{A}$ a functor.

Given a class $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$ of functorial $\mathbf{D}$ -invariants, the following cannot be satisfied simultaneously:

- ▶ $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$ is non-empty;
- ▶ U is coherent and admits a complement structure $(-)^c$ relatively to $\mathcal{I}_{\mathbf{D}}(\mathbf{C})$
- ▶ the invariant defined by U is non-nilpotent.

Applications?

► Isomorphism problems;

- ▶ Isomorphism problems;
- ▶ Graph Reconstruction Conjecture;

- ▶ Isomorphism problems;
- ▶ Graph Reconstruction Conjecture;
- ▶ $\vdots$

Thanks